I- General discrete probability

a) Variables

Definition

A random variable represents in number form the possible outcomes which could occur for some random experiment.

Definition

A discrete random variable X has a set of distinct possible values. In this course you will consider only a finite number of outcomes, so we label them $x_1, x_2, x_3, \ldots, x_n$.

Note. *To determine the value of a discrete random variable we need to count.*

Example

X could be :

— the number of houses in your suburb which have a 'power safety switch'

— the number of new bicycles sold each year by a bicycle store

— the number of defective light bulbs in the purchase order of a city store.

Definition

A continuous random variable X could take possible values in some interval on the number line.

Example

For example, X could be :

— the heights of men, which would all lie in the interval $50 < X < 250$ cm

— the volume of water in a rainwater tank, which could lie in the interval $0 < X < 100$ m^3.

Note. *To determine the value of a continuous random variable we need to measure.*

EXERCISE 1 Classify the following random variables as continuous or discrete :

1. the quantity of fat in a sausage
2. the mark out of 50 for a geography test
3. the weight of a seventeen year old student
4. the volume of water in a cup of coffee
5. the number of trout in a lake
6. the number of hairs on a cat
7. the length of hairs on a horse

EXERCISE 2 A supermarket has four checkouts A, B, C, and D. Management checks the weighing devices at each checkout. If a weighing device is accurate a Y is recorded ; otherwise, N is recorded. The random variable being considered is the number of weighing devices which are accurate.

1. Suppose X is the random variable. What values can X have ?
2. Tabulate the possible outcomes and the corresponding values for X.
3. Describe, using X, the events of :
   (a) exactly two devices being accurate
   (b) at least two devices being accurate.

b) Distribution

Definition

For any random variable there is a corresponding probability distribution which describes the probability that the variable will take any particular value.
The probability that the variable X takes value x is denoted $\mathrm{P}(X = x)$.

Example

When tossing two coins, the random variable X could be 0 heads, 1 head, or 2 heads, so $X = 0$, 1, or 2.
The associated probability distribution is $\mathrm{P}(X = 0) = \frac{1}{4}$, $\mathrm{P}(X = 1) = \frac{1}{2}$, and $\mathrm{P}(X = 2) = \frac{1}{4}$.

Example

A supermarket has three checkout points A, B, and C. A government inspector checks the weighing scales for accuracy at each checkout. If a weighing scale is accurate then Y is recorded, and if not, N is recorded. Suppose the random variable X is the number of accurate weighing scales at the supermarket.

1. List the possible outcomes.
2. Describe, using X, the events of there being :
   (a) one accurate scale
   (b) at least one accurate scale.

Property

If X is a random variable with sample space $\{x_1, x_2, x_3, \ldots, x_n\}$ and corresponding probabilities $\{p_1, p_2, p_3, \ldots, p_n\}$ so that $\mathrm{P}(X = x_i) = p_i$, $i = 1, \ldots, n$, then :

— $0 \leqslant p_i \leqslant 1$ for all $i = 1$ to n

— $\sum_{i=1}^{n} p_i = p_1 + p_2 + p_3 + \ldots + p_n = 1$

— $\{p_1, \ldots, p_n\}$ describes the **probability distribution** of X.

Example

When a coin is tossed twice, the number of heads X has sample space $\{0, 1, 2\}$ with corresponding probabilities $\left\{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right\}$.

We see that $0 \leqslant p_i \leqslant 1$ for each value of i, and that the probabilities add up to 1.

Note.

Notation	*Statement*
$\mathrm{P}(X = 3)$	the probability that X equals 3
$\mathrm{P}(X < 3)$	the probability that X is less than 3
$\mathrm{P}(X \leqslant 3)$	the probability that X is at most 3
$\mathrm{P}(X > 3)$	the probability that X is more than 3
$\mathrm{P}(X \geqslant 3)$	the probability that X is at least 3
$\mathrm{P}(3 < X < 7)$	the probability that X is between 3 and 7
$\mathrm{P}(3 \leqslant X \leqslant 7)$	the probability that X is at least 3 but no more than 7
$\mathrm{P}(3 < X \leqslant 7)$	the probability that X is more than 3 but no more than 7
$\mathrm{P}(3 \leqslant X < 7)$	the probability that X is at least 3 but less than 7

Example

A magazine store recorded the number of magazines purchased by its customers in one week. 23% purchased one magazine, 38% purchased two, 21% purchased three, 13% purchased four, and 5% purchased five.

1. What is the random variable ?
2. Make a probability table for the random variable.
3. Graph the probability distribution using a spike graph.

Example

Show that the following are probability distribution functions :

1. $\mathrm{P}(x) = \dfrac{x^2+1}{34}, \quad x = 1, 2, 3, 4$
2. $\mathrm{P}(x) = \binom{3}{x}(0.6)^x(0.4)^{3-x}, \quad x = 0, 1, 2, 3$

EXERCISE 3 Find k in these probability distributions :

1.

x	0	1	2
$\mathrm{P}(X = x)$	0.3	k	0.5

2.

x	0	1	2	3
$\mathrm{P}(X = x)$	k	$2k$	$3k$	k

EXERCISE 4 Sally's number of hits in each softball match has the following probability distribution :

x	0	1	2	3	4	5
$\mathrm{P}(X = x)$	0.07	0.14	k	0.46	0.08	0.02

1. State clearly what the random variable represents.
2. Find k.
3. Find : **i** $\mathrm{P}(X \geqslant 2)$ **ii** $\mathrm{P}(1 \leqslant X \leqslant 3)$

Example

Two marbles are randomly selected without replacement from a bag containing 4 red and 2 blue marbles. Let X denote the number of red marbles selected.

1. Find the probability distribution of X.
2. Illustrate the probability distribution using a spike graph.

EXERCISE 5 Electrical components are produced and packed into boxes of 10. It is known that 4% of the components produced are faulty. The random variable X denotes the number of faulty items in the box, and has probability distribution function

$$\mathrm{P}(X = x) = \binom{10}{x}(0.04)^x(0.96)^{10-x}, \quad x = 0, 1, 2, \ldots, 10.$$

1. Find the probability that a randomly selected box will not contain a faulty component.
2. Find the probability that a randomly selected box will contain at least one faulty component.

EXERCISE 6 A bag contains 5 blue and 3 green tickets and a number of tickets are randomly selected without replacement. We let X denote the number of blue tickets selected. Find the probability distribution of X if :

1. two tickets are randomly selected
2. three tickets are randomly selected.

EXERCISE 7 When a pair of dice is rolled, D denotes the sum of the top faces.

1. Display the possible results in a table.
2. Find $\mathrm{P}(D = 7)$.
3. Find the probability distribution of D.
4. Find $\mathrm{P}(D \geqslant 8 \mid D \geqslant 6)$.

EXERCISE 8 When a pair of dice is rolled, N denotes the difference between the numbers on the top faces.

1. Display the possible results in a table.
2. Construct a probability distribution table for the possible values of N.
3. Find $\mathrm{P}(N = 3)$.
4. Find $\mathrm{P}(N \geqslant 3 \mid N \geqslant 1)$.

c) Expectation

Property

If there are n trials of an experiment, and an event has probability p of occurring in each of the trials, then the number of times we expect the event to occur is np.

Property

The expected outcome for the random variable X is the mean result μ.
In general, the expectation of the random variable X is

$$\mathbf{E}(X) = \mu = \sum_{i=1}^{n} x_i p_i \qquad \text{or} \qquad \sum_{i=1}^{n} x_i \,\mathrm{P}(X = x_i)$$

Example

Each time a footballer kicks for goal he has a $\frac{3}{4}$ chance of being successful. In a particular game he has 12 kicks for goal. How many goals would you expect him to kick ?

Property

Suppose X represents the gain of a player from each game. The game is fair if $\mathrm{E}(X) = 0$.

Example

In a game of chance, a player spins a square spinner labelled 1, 2, 3, 4. The player wins the amount of money shown in the table alongside, depending on which number comes up. Determine :

1. the expected return for one spin of the spinner
2. the expected gain of the player if it costs \$5 to play each game
3. whether you would recommend playing this game.

Number	1	2	3	4
Winnings	\$1	\$2	\$5	\$8

EXERCISE 9

1. If 3 coins are tossed, what is the chance that they all fall heads ?
2. If the 3 coins are tossed 200 times, on how many occasions would you expect them all to fall heads ?

EXERCISE 10 A single coin is tossed once. If a head appears you win \$2, and if a tail appears you lose \$1. How much would you expect to win when playing this game three times ?

EXERCISE 11 During the snow season there is a $\frac{3}{7}$ probability of snow falling on any particular day. If Udo skis for five weeks, on how many days could he expect to see snow falling ?

EXERCISE 12 In a random survey of her electorate, politician A discovered the residents' voting intentions in relation to herself and her two opponents B and C. The results are indicated alongside :

A	B	C
165	87	48

1. Estimate the probability that a randomly chosen voter in the electorate will vote for :
   (a) A
   (b) B
   (c) C.
2. If there are 7500 people in the electorate, how many of these would you expect to vote for :
   (a) A
   (b) B
   (c) C ?

EXERCISE 13 A person rolls a normal six-sided die and wins the number of euros shown on the face.

1. Find the expected return from one roll of the die.
2. Find the expected *gain* of the player if it costs 4 to play the game. Would you advise the person to play several games ?
3. Suppose it costs k to play the game. What value(s) of k will result in :
   (a) a fair game
   (b) a profit being made by the vendor ?

EXERCISE 14 At a charity event there is a money-raising game involving a pair of ordinary dice. The game costs $\$a$ to play. When the two dice are rolled, their sum is described by the variable X.
The organisers decide that a sum which is less than 4 or between 7 and 9 inclusive is a loss of $\$\frac{a}{3}$, a result between 4 and 6 inclusive gives a return of \$7, and a result of 10 or more gives a return of \$21.

1. Determine P$(X \leqslant 3)$, P$(4 \leqslant X \leqslant 6)$, P$(7 \leqslant X \leqslant 9)$, and P$(X \geqslant 10)$.
2. Show that the expected gain of a player is given by $\frac{1}{6}(35 - 7a)$ dollars.
3. What value would a need to have for the game to be 'fair' ?
4. Explain why the organisers would not let a be 4.
5. If the organisers let a be 6 and the game was played 2406 times, estimate the amount of money raised by this game.

II- Binomial Distribution

We now examine a special type of discrete random variable which is applied to sampling with replacement. The probability distribution associated with this variable is the **binomial probability distribution**.

a) Experiments

Definition

Consider an experiment for which there are two possible results : **success** if some event occurs, or **failure** if the event does not occur.
If we repeat this experiment in a number of **independent trials**, we call it a **binomial experiment**.

Property

The probability of a success p must be constant for all trials. Since success and failure are complementary events, the probability of a failure is $1-p$ and is constant for all trials.

Note. *The random variable X is the total number of successes in n trials.*

Example

1. Expand $\left(\frac{9}{10}+\frac{1}{10}\right)^5$.
2. An archer has a 90% chance of hitting a target with each arrow. If 5 arrows are fired, determine the chance of hitting the target :
   (a) twice only
   (b) at most 3 times.

EXERCISE 15

1. Expand $(p+q)^5$.
2. If *five* coins are tossed simultaneously, what is the probability of getting :
   (a) 4 heads and 1 tail in any order
   (b) 2 heads and 3 tails in any order
   (c) 4 heads and 1 tail in that order ?

EXERCISE 16

1. Expand $\left(\frac{2}{3}+\frac{1}{3}\right)^4$.
2. Four chocolates are selected at random, with replacement, from a box which contains strawberry creams and almond centres in the ratio $2:1$. Find the probability of getting :
   (a) all strawberry creams
   (b) two of each type
   (c) at least 2 strawberry creams.

EXERCISE 17

1. Expand $\left(\frac{3}{4}+\frac{1}{4}\right)^5$.
2. In New Zealand in 1946 there were two different coins of value two shillings. These were 'normal' kiwis and 'flat back' kiwis, in the ratio $3:1$. From a very large batch of 1946 two shilling coins, five were selected at random with replacement. Find the probability that :
   (a) two were 'flat backs'
   (b) at least 3 were 'flat backs'
   (c) at most 3 were 'normal' kiwis.

b) Binomial probability distribution function

Property

Consider a binomial experiment for which p is the probability of a *success* and $1-p$ is the probability of a *failure*.
If there are n independent trials then the probability that there are r *successes* and $n-r$ *failures* is

$$\mathrm{P}(X=r)=\binom{n}{r}p^r(1-p)^{n-r} \quad \text{where} \quad r=0,1,2,3,4,\ldots,n.$$

$\mathrm{P}(X=r)$ is the **binomial probability distribution function**.

Property

The expected or mean outcome of the experiment is $\mu = \mathrm{E}(X) = np$

If X is the random variable of a binomial experiment with parameters n and p, then we write $\boldsymbol{X \sim \mathbf{B}(n,\, p)}$ where $\sim$ reads "*is distributed as*".

Example

72% of union members are in favour of a certain change to their conditions of employment. A random sample of five members is taken. Find :

1. the probability that three members are in favour of the change in conditions
2. the probability that at least three members are in favour of the changed conditions
3. the expected number of members in the sample that are in favour of the change.

Example

On the GDC :

— To find the probability $\mathrm{P}(X = r)$ that the variable takes the value r, we use the **binomial probability distribution function**.

— To find the probability that the variable takes a *range* of values, such as $\mathrm{P}(X \leqslant r)$ or $\mathrm{P}(X \geqslant r)$, we use the **binomial cumulative distribution function**.

EXERCISE 18 At a manufacturing plant, 35% of the employees work night-shift. If 7 employees are each selected from the entire group at random, find the probability that :

1. exactly 3 of them work night-shift
2. less than 4 of them work night-shift
3. at least 4 of them work night-shift.

EXERCISE 19 There is a 5% chance that any apple in a crate will have a blemish. If a random sample of 25 apples is taken with replacement, find :

1. the probability that exactly 2 of these have blemishes
2. the probability that at least one has a blemish
3. the expected number of apples that will have a blemish.

EXERCISE 20 During a season, a basketball player has a 94% success rate in shooting from the free throw line. In one match the basketballer has 20 shots from the free throw line.

1. Find the probability that the basketballer is successful on :
   (a) all 20 throws
   (b) at least 18 throws.
2. Find the expected number of successful throws for the match.

EXERCISE 21 If a fair coin is tossed 200 times, find the probability that :

1. between 90 and 110 (inclusive) heads turn up
2. more than 95 but less than 105 heads turn up.

c) Mean and standard deviation

Property

Suppose X is a binomial random variable with parameters n and p, so $X \sim \text{B}(n, p)$.

— The **mean** of X is $\boldsymbol{\mu = np}$.

— The **standard deviation** of X is $\boldsymbol{\sigma = \sqrt{np(1-p)}}$.

— The **variance** of X is $\boldsymbol{\sigma^2 = np(1-p)}$.

Example

A fair die is rolled twelve times and X is the number of sixes that could result. Find the mean and standard deviation of the X-distribution.

Example

5% of a batch of batteries are defective. A random sample of 80 batteries is taken with replacement. Find the mean and standard deviation of the number of defective batteries in the sample.

EXERCISE 22 Bolts produced by a machine vary in quality. The probability that a given bolt is defective is 0.04. Random samples of 30 bolts are taken from the week's production.

1. If X is the number of defective bolts in a sample, find the mean and standard deviation of the X-distribution.
2. If Y is the number of non-defective bolts in a sample, find the mean and standard deviation of the Y-distribution.

EXERCISE 23 A city restaurant knows that 13% of reservations are not honoured, which means the group does not arrive. Suppose the restaurant receives 30 reservations. Let X be the random variable of the number of groups that do not arrive. Find the mean and standard deviation of the X-distribution.